The MapEval4OceanHeat experimental protocol

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Version updated starting on: 09.07.23

Knowledge of the spatio-temporal evolution of ocean heat content (OHC) is essential to our understanding of both past and future climate change. At global scales, ocean heat uptake is a key element of the global energy and sea level budgets and provides a powerful constraint on climate sensitivity and past climate forcings. At regional scales, OHC changes are important to our understanding of climate variability and for the development, initialization, and evaluation of seasonal-to-decadal prediction systems. However, international intercomparison efforts have repeatedly shown large spread in both global OHC change and the spatial patterns of change over the past decades estimated by various research groups. Even for the relatively well-observed 0-700 m ocean, the uncertainty introduced by the choice of mapping method is large.

Building on previous intercomparison efforts and motivated by successful “proof-of-concept” studies, the MapEval4OceanHeat initiative will carry out an objective assessment of mapping methods based on “synthetic” ocean observations from high-resolution climate and ocean modelling systems. Through collaboration with the international research community, MapEval4OceanHeat will evaluate the strengths and limitations of contemporary mapping approaches and assess our ability to constrain OHC change at global and regional scales.

This document summarizes the experimental protocol for the intercomparison of mapping methods based on “synthetic profiles” (Figure 1). The synthetic profiles are model analogues of the real-world EN4 temperature profile observations (https://www.metoffice.gov.uk/hadobs/en4) that have been extracted from a 1/10th degree ocean model simulation. The plan is that different mapping methods are used to interpolate the synthetic profiles for a given depth layer and the results inter-compared to further our understanding of the different capabilities/limitations of each method, including the associated uncertainties (where available). The experimental design is similar to the study published by Allison et al (2019): https://iopscience.iop.org/article/10.1088/1748-9326/ab2b0b

Ultimately, a series of monthly 2D maps of ocean heat content anomaly produced by each mapping method will be compared to the model “truth” field (this field will be withheld until after the mapping exercise has been completed). The following text outlines the technical details of the protocol, including specifying the preferred data format for receiving the mapped ocean heat content anomaly fields. The protocol itself is “tiered”, meaning that there are a variety of options to the basis of which the mapping can be conducted (see experiments A-E). This is to promote flexibility and inclusion of the largest possible number of mapping methods.

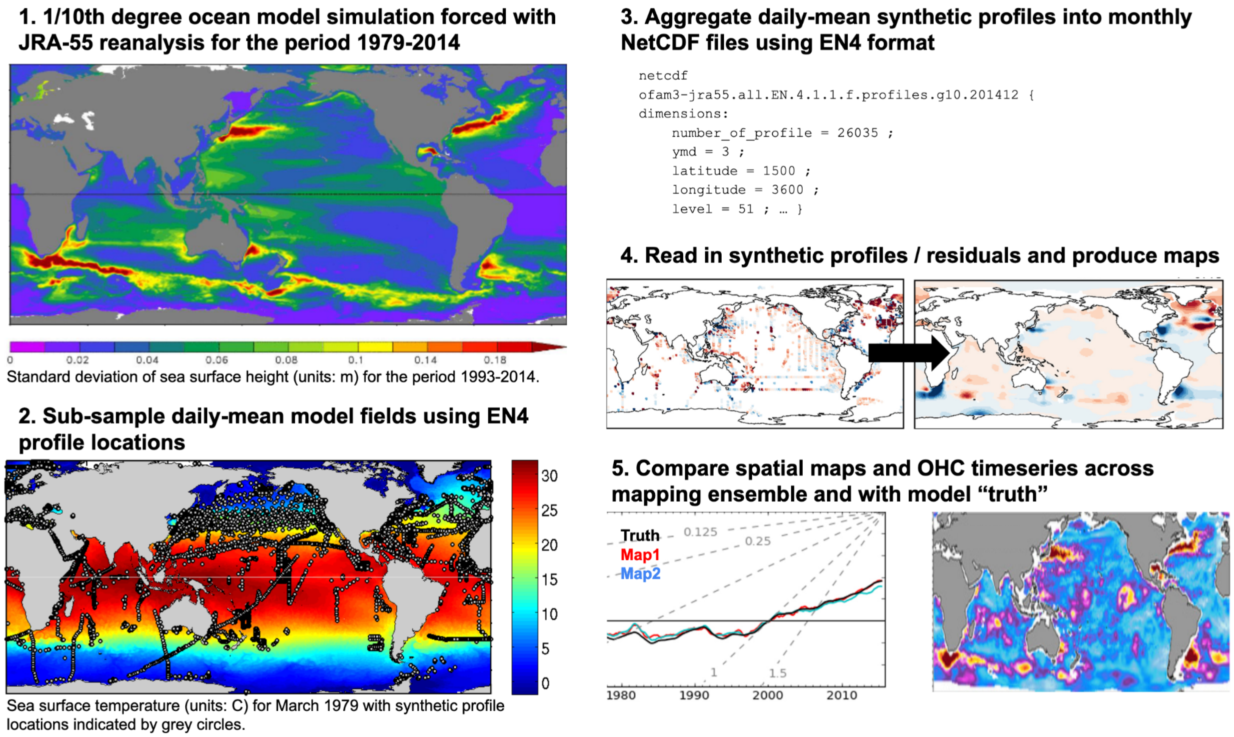


Figure 1: Schematic of the ME4OH experimental protocol work-flow.

Definitions to read this document:

- OHC point data: ocean heat content synthetic data (vertically integrated temperature profiles, multiplied by cp0 and rho0)
    - cp0 = 3989.244 J/kg/K is heat capacity
    - rho0= 1030 kg/m3 is density
- Time period
    - T1: years 1979-2014 (whole period)
    - T2: years 2005-2014 (Argo period)
    - T3: years 1993-2014 (satellite period)
- Residuals: total field minus the selected climatology.
- Climatology
    - C1: a climatology that describes the seasonal cycle and linear trends, e.g. given a depth layer of interest, C1_T1 is C1(x,y,t) computed for the years in T1. C1 on 1 January, 2010 <u>is not</u> the same as C1 on 1 January, 2011, as C1 includes a linear trend. Timeseries of residuals computed using C1 are close to stationary.
    - C2: a climatology that describes only the seasonal cycle, e.g. given a depth layer of interest, C2_T1 is C2(x,y,t) computed for the years in T1 and it changes in time only due to the seasonal cycle. C2 on 1 January, 2010 <u>is</u> the same as C2 on 1 January, 2011, as C2 <u>does not</u> include a linear trend. Timeseries of residuals computed using C2 are NOT stationary.
- Vertical levels selected to compute OHC point data:

  - L1: 0-286.6 m layer
  - L2: 286.6-685.9 m layer (i.e. starting from the lower bound of layer 286.6m, i.e. not including the layer associated with nominal model depth 286.6m)
  - L3: 685.9-1985.3 m layer (i.e. starting from the lower bound of layer 685.9, i.e. not including the layer associated with nominal model depth 685.9m)

Please note that the nominal model depths are used here to identify the levels (L1, L2 and L3) as these are the depths where temperature and salinity are reported. The OHC integration (**see section on "Vertical integration" later in this document**) uses the depth cell thicknesses with lower bounds deeper than the nominal depth, hence L1 integrates OHC down to 306.25m, L2 is the integral between 306.25m and 730.93m, and L3 between 730.93m and 2075.25m.

Experimental design:

- **Experiment A*:** starting from residuals for OHC point data in the layers L1, L2, L3 using C1, either for T1 or T2. If possible, repeat for C2.
(rationale: most cleanly isolates the mapping step)

- **Experiment B*:** starting from OHC point data in the layers L1, L2, L3, either for T1 or T2 (groups will estimate their own climatology). (rationale: may be easy to leverage existing workflows; also, the integral is computed with the same method as the model truth; the comparison with experiment A will provide insights on how well the climatology is estimated)

- **Experiment C*:** starting from full profiles of potential temperature, with no climatology removed. Groups will need to estimate the climatology or use one that is already available. Please use cp0 and rho0 listed in the previous section to estimate OHC. Also, if possible, we encourage groups to do the vertical integrals using the same method in section "Vertical integral" (in the following), for consistency with the model truth. (rationale: easiest to leverage existing workflows)

- **Experiment D:** starting from residuals for model profiles in the layers L1, L2, L3 using C1, either for T1 or T2. Please use cp0 and rho0 listed in the previous section and see the section "Vertical integral" as for experiment C. If possible, repeat for C2.

- **Experiment E:** other (e.g. could use SSTA and SSHA model fields on the model grid)

*In the interest of time, we will start with experiment B and C (C is for those who cannot perform experiment B) for the OSM2024 presentation in February, so that, as much as possible, maps are compared across products by different groups starting from the same input. For the manuscript, we would like to include Experiments A and D. We will reach out again about the manuscript.

Data inputs:
Input (synthetic) data are provided via the CSIRO Data Access Portal (Appendix A) with EN4 format (Appendix B), i.e. model daily profiles corresponding to the EN4 records are packed in monthly files, as with the EN4 observed profile data. The model (Ocean Forecasting Australia Model version 3, OFAM3) was forced with the Japanese 55-year Reanalysis (JRA-55) surface reanalysis fields. Since

the model data are based on daily means, the time of day for the profile is ignored. For each month (over the 1979 to 2014 period), prognostic model variables are sampled, i.e. potential temperature (temp, degree Celsius), practical salinity (salt, psu), velocity components (u, v, w; m/s) and dynamical sea surface height (eta_t; m), and stored in the same file. OHC point data (TJ/m^2) are computed and stored in the file as well. **Please read how to access input (synthetic) data for the experiments in Appendix A (links to the files and instructions on what variables to use are included in the appendix).**

Vertical integral

OHC density (TJ/m^2) is computed from the model as the <u>cumulative sum (from the ocean surface up to the nominal model depth cell lower bound of interest included) of potential temperature</u> (expressed in degrees Kelvin) multiplied by both the corresponding depth level thicknesses and cp0*rho0, see MatLab code snippet below. The depth level thicknesses are saved in the variable 'ts_dz', listed in Appendix B. As described in Appendix A, the 'dohc' variable (see NetCDF meta-data reported in Appendix B) **stores the separate contributions of layers L1, L2 and L3, i.e. <u>L1 integrates OHC down to 306.25m, L2 is the integral between 306.25m and 730.93m, and L3 between 730.93m and 2075.25m.</u>**

<u>As an example, to calculate L2, we first calculate L1 (integrating down to 306.25m) and L2_0 (integrating down to 730.93m): L2 is then equal to L2_0 minus L1.</u>

```
%%  Profile ocean heat content density ------------------------------------
%
function dohc = me4oh_dohc(temp,z)

      rho0  = 1030;      % kg/m^3
      Cp0   = 3989.244; % J/(kg K)
      T0    = 273.15;   % Celcius -> Kelvin
      tera  = 10^12;
      scale = Cp0 * rho0/tera;

%  Hereafter the variable 'z' does not store the model nominal depths (e.g. L1, L2
%  or L3), but the vertical depth cell bounds, hence 'diff(z)' equates the depth
%  cell thicknesses. I note that the depth bounds have not been stored in the
%  synthetic profile files, as the OHC integration used here only needs
%  thicknesses.

      dz    = diff(z) * scale;

%%  In the cumsum below the default behaviour is used ---------------------
%
%     Y = cumsum(___,NANFLAG) specifies how NaN (Not-A-Number) values are
%     treated:
%
%     'includenan' - (default) the sum of elements containing NaN values is NaN.
%     'omitnan'    -           the sum of elements containing NaN values is
%                              the sum of all non-NaN elements. If all
%                              elements are NaN, the result is 0.
%
```

```
    dohc  = cumsum((temp + T0) .* dz); % Km(kg/m^3)(J/kg K)/10^12 = TJ/m^2
end
%
%%  End of me4oh_dohc ------------------------------------------------------
```

Output groups are asked to share:

- If possible, please provide maps (i.e. spatially-continuous 3D arrays for each layer (x,y,t), with the format indicated in the next section) created from mapping
    - the total OHC point data we provide, i.e. output from Experiment B for layer L1, L2, L3 (for OSM 2024 in February); and
    - the residuals for OHC point data we provide, i.e. output from Experiment A for layer L1, L2, L3 (for the manuscript and, if you have time, even for OSM 2024)

  OR
    - the raw synthetic profiles we provide, i.e. output from Experiment C for layer L1, L2, L3 (for OSM 2024 in February); and
    - the raw synthetic profile residuals we provide, i.e. output from Experiment D for layer L1, L2, L3 (for the manuscript and, if you have time, even for OSM 2024).
- If that is not possible, please provide what is discussed on a case to case basis, depending on what input you are starting from.

Format of file to submit to ME4OH: One NetCDF file for each depth layer, e.g. DATA(LONGITUDE,LATITUDE,TIME), LONGITUDE(LONGITUDE), LATITUDE(LATITUDE), TIME(TIME); TIME should be in days since 1-JAN-1900 00:00:00. If you are unable to comply with the data format specifications, please contact us to discuss further. The name of the file should indicate the depth level, the time period, the experiment, and the product name, i.e. OHC_2005_2014_lev0_286.6_expA_productNAME.nc (e.g. productNAME could be RoemmichGilson)

Time resolution of the maps: monthly or, if monthly is not possible, yearly (to be intended as the average over the full year)

Spatial resolution of the maps: Preferred grid is a standard latitude-longitude grid of 1x1 degrees using lon (20.5 to 379.5) and lat (-89.5 to 89.5), so that no interpolation is needed for the intercomparison. We can accommodate alternative grids if absolutely necessary, please let us know if this is the case for your mapping method.

Note on QC of input data: all the data we provide (and that are not NaNs) should be used (there should be no additional layer of QC applied to the input data).

Note about output maps: We will compare products based on a common ocean_mask: in each product, all points that are not-NaN (at all timesteps) will be used to create the mask.

Info groups should provide along with their output (via a Google form):

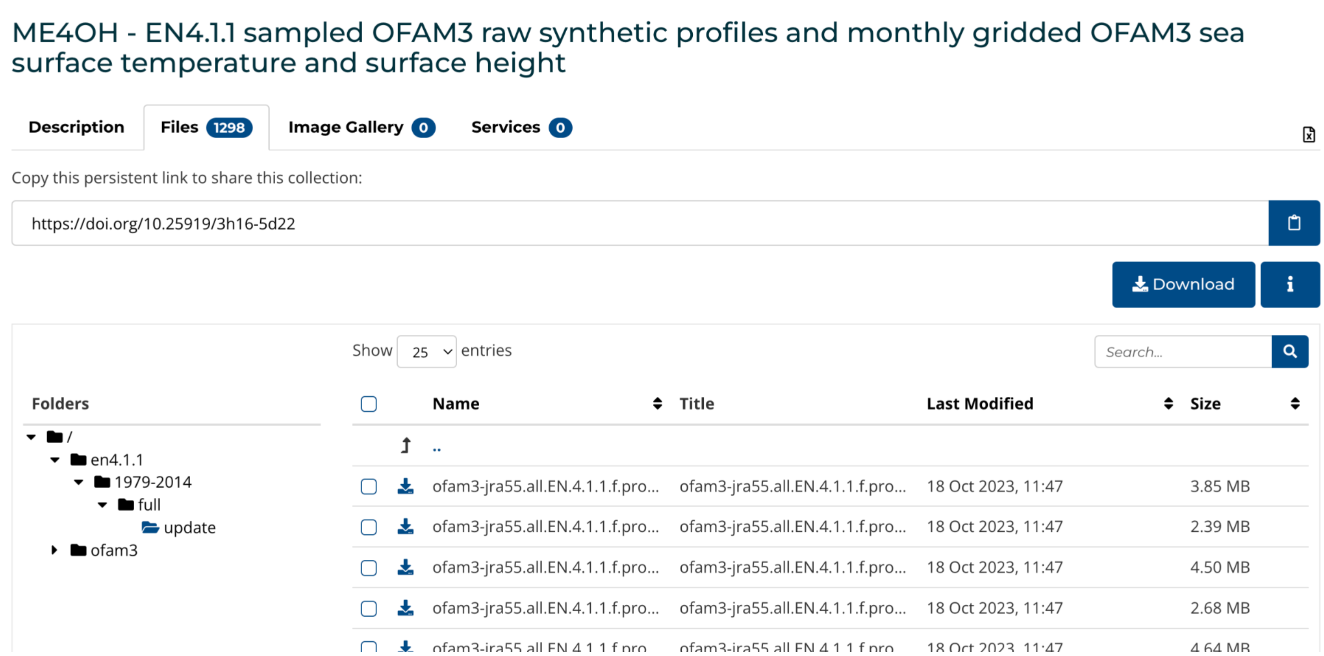
- Did you focus on T1 or T2?
- Did you use C1 or C2?
- Which experiments did you run and are providing us with output for? (if experiment E, please provide details; this information is necessary to understand the comparison across products)

- Did you change anything from the default protocol? (this information is necessary to understand the comparison across products)
- Please provide a paragraph describing your method including a citation, if available

Appendix A: How to access input files and what variables to use for the experiments

To access the synthetic profiles, please use this link. You will land in the main page for the data collection, e.g.

Please click on "Files 1298" (as seen on the top left of the screenshot above). Then use the manu on the left (see "Folders" on the middle left in the screenshot below) to navigate to the synthetic data of interest in the folder "update", e.g.



An example file name for the files of interest is ofam3-jra55.all.EN.4.1.1.f.profiles.g10.197901.update.nc (please note file names ending in *update.nonan.nc should **not** be used for our project). Duplicates (which could occur, as the model is daily) and dry points (i.e locations where EN4.1.1 was wet but the model dry) have been removed. In the files shared (see metadata in Appendix B), the variable **dohc** (TJ/m^2) stores ocean heat content

(OHC) density on 3 selected model depth levels: L1, L2, L3, (These depth levels are commonly used to report OHC from observations.) Files may have NaNs for a certain layer when the bathymetry at a given location is shallower than the selected depth. **As an example**, for layer L1 experiment B during the Argo period, you would start from point data in dohc(:,level_bound=1) combining these point data across all the .nc files for months during 2005-2014. **Please also note**:

- longitude to use for the experiments is stored in the variable **ts_lon**, with values 0-360
- latitude is in **ts_lat**
- time is in **en4_ymd** and en4_ymd(1,:) is the year, en4_ymd(2,:) is the month, and en4_ymd(3,:) is the day (reminder: the model is daily, hence there is no hour, min, sec info)
- vertical levels (if not using dohc) are in **ts_z**(level)
- depth level thicknesses (if not using dohc) are saved in the variable **ts_dz**
- temperature profiles** (if not using dohc) are in **temp**(number_of_profile, level)
- ssh (if needed) is in **eta_t**(number_of_profile)
- sst (if needed) is in **sst**(number_of_profile)
- salinity profiles (if needed) are in **salt**(number_of_profile, level)
- other variables in the files are not needed for our project

As some methods also need gridded SST and SSH, we will provide information about how to access these, to groups that request them. These files have a format as illustrated in Appendix C and D.

While we agree conservative temperature is a more relevant quantity for OHC, for the purpose of ME4OH experiments we will use potential temperature for input data and model truth. dohc in the files shared is computed based on potential temperature from the model (the model was using EOS-80 sea water equation), hence, if you use profile data (instead of dohc) please do not convert the profiles we provide to conservative temperature (or other quantities).

We believe our results from (model) potential temperature will still be helpful to assess mapping methods and their uncertainties. As always, please let us know if you have any questions/comments.

Appendix B: Updated EN-4.1.1 sampled OFAM3 raw synthetic profile files

```
ofam3-jra55.all.EN.4.1.1.f.profiles.g10.201412.update {
Dimensions:
	number_of_profile = 20309 ;
	ymd = 3 ;
	level = 51 ;
	level_bound = 3 ;
	nv = 2 ;
Variables:
	float en4_lat(number_of_profile) ;
		en4_lat:units = "degree North" ;
	float en4_lon(number_of_profile) ;
		en4_lon:units = "degree East" ;
	float en4_ymd(number_of_profile, ymd) ;
		en4_ymd:units = "[year month day-of-month]" ;
	float en4_ydec(number_of_profile) ;
		en4_ydec:units = "decimal year" ;
	float ts_lat(number_of_profile) ;
```

```
            ts_lat:units = "degree North" ;
      float ts_lon(number_of_profile) ;
            ts_lon:units = "degree East" ;
      float ts_z(level) ;
            ts_z:units = "m" ;
            ts_z:short_name = "depth_level" ;
      float ts_dz(level) ;
            ts_dz:units = "m" ;
            ts_dz:short_name = "depth_level_thickness" ;
      float ts_bound(nv, level_bound) ;
            ts_bound:units = "m" ;
            ts_bound:short_name = "ohc_depth_bound" ;
      float eta_t(number_of_profile) ;
            eta_t:units = "m" ;
            eta_t:short_name = "dynamic_sea_level" ;
      float sst(number_of_profile) ;
            sst:units = "degree Celcius" ;
            sst:short_name = "sea_surface_temperature" ;
      float dohc(number_of_profile, level_bound) ;
            dohc:units = "TJ/m^2" ;
            dohc:short_name = "ocean_heat_content_density" ;
      float temp(number_of_profile, level) ;
            temp:units = "degree Celcius" ;
            temp:short_name = "potential_tempareture" ;
      float salt(number_of_profile, level) ;
            salt:units = "psu" ;
            salt:short_name = "practical_salinity" ;
// global attributes:
      :creation_date = "22-Sep-2023 09:47:07" ;
      :creation_by = "me4oh_en411_synprofile_netcdf_write" ;
      :en4_source =
      "/g/data/xv83/dpm599/OFAM3/matlab/ofam3_sampling/nc/en4.1.1/ofam3-jra5
      :ofam_source = "..." ;
}
```

Appendix C: OFAM3 sea surface temperature

```
temp_ofam3_month_197901-201412 {
Dimensions:
      Time = UNLIMITED ; // (432 currently)
      bnds = 2 ;
      xt_ocean = 3600 ;
      yt_ocean = 1500 ;
      st_ocean = 1 ;
Variables:
      double Time(Time) ;
            Time:standard_name = "time" ;
            Time:long_name = "Time" ;
            Time:bounds = "Time_bnds" ;
            Time:units = "days since 1979-01-01 00:00:00" ;
            Time:calendar = "gregorian" ;
```

```
            Time:axis = "T" ;
      double Time_bnds(Time, bnds) ;
      double xt_ocean(xt_ocean) ;
            xt_ocean:standard_name = "longitude" ;
            xt_ocean:long_name = "tcell longitude" ;
            xt_ocean:units = "degrees_E" ;
            xt_ocean:axis = "X" ;
      double yt_ocean(yt_ocean) ;
            yt_ocean:standard_name = "latitude" ;
            yt_ocean:long_name = "tcell latitude" ;
            yt_ocean:units = "degrees_N" ;
            yt_ocean:axis = "Y" ;
      double st_ocean(st_ocean) ;
            st_ocean:long_name = "tcell zstar depth" ;
            st_ocean:units = "meters" ;
            st_ocean:positive = "down" ;
            st_ocean:axis = "Z" ;
            st_ocean:cartesian_axis = "Z" ;
            st_ocean:edges = "st_edges_ocean" ;
      double average_DT(Time) ;
            average_DT:long_name = "Length of average period" ;
            average_DT:units = "days" ;
            average_DT:cell_methods = "Time: mean" ;
      short temp(Time, st_ocean, yt_ocean, xt_ocean) ;
            temp:standard_name = "sea_water_potential_temperature" ;
            temp:long_name = "Potential temperature" ;
            temp:units = "degrees C" ;
            temp:add_offset = 45.f ;
            temp:scale_factor = 0.001678518f ;
            temp:_FillValue = -32768s ;
            temp:missing_value = -32768s ;
            temp:cell_methods = "time: mean" ;
            temp:packing = 4 ;
            temp:time_avg_info = "average_T1,average_T2,average_DT" ;
// global attributes:
      :CDI = "Climate Data Interface version 2.0.5 (https://mpimet.mpg.de/cdi)" ;
      :Conventions = "CF-1.6" ;
      :history = "..." ;
      :filename = "TMP/ocean_ofam_1979_01.nc.0000" ;
      :NumFilesInSet = 720 ;
      :title = "jra_55_1979" ;
      :grid_type = "regular" ;
      :NCO = "4.3.8" ;
      :frequency = "mon" ;
      :CDO = "Climate Data Operators version 2.0.5 (https://mpimet.mpg.de/cdo)" ;
}
```

APPENDIX D: OFAM3 sea surface height - dynamic sea-level

```
etat_ofam3_month_197901-201412 {
Dimensions:
```

```
	yt_ocean = 1500 ;
	Time = UNLIMITED ; // (432 currently)
	xt_ocean = 3600 ;
Variables:
	double yt_ocean(yt_ocean) ;
		yt_ocean:long_name = "tcell latitude" ;
		yt_ocean:units = "degrees_N" ;
		yt_ocean:cartesian_axis = "Y" ;
	double Time(Time) ;
		Time:long_name = "Time" ;
		Time:units = "days since 1979-01-01 00:00:00" ;
		Time:cartesian_axis = "T" ;
		Time:calendar_type = "GREGORIAN" ;
		Time:calendar = "GREGORIAN" ;
		Time:bounds = "Time_bounds" ;
	double xt_ocean(xt_ocean) ;
		xt_ocean:long_name = "tcell longitude" ;
		xt_ocean:units = "degrees_E" ;
		xt_ocean:cartesian_axis = "X" ;
	float eta_t(Time, yt_ocean, xt_ocean) ;
		eta_t:long_name = "surface height on T cells [Boussinesq (volume
	conserving) model]"
		eta_t:units = "meter" ;
		eta_t:valid_range = -10.f, 10.f ;
		eta_t:missing_value = -1.e+20f ;
		eta_t:_FillValue = -1.e+20f ;
		eta_t:cell_methods = "time: mean" ;
		eta_t:time_avg_info = "average_T1,average_T2,average_DT" ;
		eta_t:coordinates = "geolon_t geolat_t" ;
// global attributes:
	:filename = "ocean_month_1979_01.nc" ;
	:title = "jra_55_1979" ;
	:grid_type = "regular" ;
	:grid_tile = "N/A" ;
	:history = "..." ;
	:NCO = "\"4.6.4\"" ;
	:nco_openmp_thread_number = 1 ;
}
```